



Modulation of lipopolysaccharide-induced proinflammatory cytokine production in vitro and in vivo by the herbal constituents apigenin (chamomile), ginsenoside Rb₁ (ginseng) and parthenolide (feverfew)

Alexa T. Smolinski^{a,b}, James J. Pestka^{a,b,c,*}

^aDepartment of Food Science and Human Nutrition, Michigan State University, 234 G.M. Trout Food Science Bldg, East Lansing, MI 48824-1224, USA

^bInstitute for Environmental Toxicology, Michigan State University, East Lansing, Michigan, USA

^cDepartment of Microbiology and Molecular Genetics, Michigan State University, East Lansing, Michigan, USA

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Abstract

Dietary supplements are not subject to the same pre-market approval as conventional drugs, thus the true efficacy and, in cases, safety of these products is not known. The objective of this study was to evaluate the potential anti-inflammatory properties of three herbal constituents, apigenin (chamomile), ginsenoside Rb₁ (ginseng) and parthenolide (feverfew) on lipopolysaccharide (LPS)-induced proinflammatory cytokine production, and to determine if effects in cell culture could predict results in an intact animal model. Murine macrophage cells and mice were treated with the stimulant LPS and herbal constituents, and resultant culture supernatant and serum, respectively, were evaluated for interleukin (IL)-6 and tumor necrosis factor (TNF)- α by ELISA. All three constituents inhibited LPS-induced IL-6 and/or TNF- α production in culture. Inhibition of these two cytokines was observed in mice, but did not display the same patterns of inhibition as cell culture data. The results suggest that all three constituents possessed anti-inflammatory properties, but that cell culture data can only be used to approximate potential effects in animals, and must be confirmed using appropriate animal models.

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1. Introduction

Hundreds of herbal remedies have been used historically in maintenance of health and in the treatment of diseases. Today, such alternatives to over-the-counter medicines and prescriptions are still sought, with 36% of the US population using herbal products (McVean et

al., 2000; O'Hara et al., 1998). The herbal supplement industry experienced tremendous growth throughout the 1990s with sales doubling every four years (Fleming, 1998) and projected sales expected to reach \$6.6 billion by 2003 (Sloan, 2000). The overwhelming increase in herbal supplement sales has prompted concern over the need for appropriate regulations. Herbal supplements are legally defined as dietary supplements, and thus fall under the regulation of the Dietary Supplement Health and Education Act of 1994 (DSHEA). Dietary supplements, unlike conventional drugs, do not require approval from the Food and Drug Administration (FDA) prior to marketing and sale of the product. Therefore, no investigations of the safety and efficacy of herbal products are required. Thus, it is often not known if these products are effective or even safe.

Abbreviations: alpha, α ; minute, min; hour, h; DSHEA, Dietary Supplement Health & Education Act; FDA, Food & Drug Administration; LPS, lipopolysaccharide; TNF, tumor necrosis factor; IL, interleukin; COX, cyclooxygenase; PG, prostaglandin; ICAM, intercellular adhesion molecule; DMSO, dimethyl sulfoxide; DMEM, Dulbecco's modified Eagle's medium; FBS, fetal bovine serum; ATCC, American Type Culture Collection; LT, leukotriene.

* Corresponding author. Tel.: +1-517-353-1709; fax: +1-517-353-8963.

E-mail address: pestka@msu.edu (J.J. Pestka).

One potential use for herbals is the treatment of inflammatory conditions. A number of inflammatory mediators are released by cells in response to localized injury or trauma. These mediators elicit or enhance particular functions of the inflammatory response and may be monitored to assess an inflammatory response. The primary inflammatory mediators are grouped into four main categories: chemokine, plasma enzyme, lipid and cytokine. Three herbal constituents, apigenin (chamomile), ginsenoside Rb₁ (ginseng) and parthenolide (feverfew), may possess anti-inflammatory properties. A limited amount of information now exists relating these herbal products, and their constituents, to specific anti-inflammatory mechanisms.

German chamomile, derived from *Matricaria recutita*, has been traditionally used as a popular folk remedy to treat digestive disorders, cramps, various skin conditions and minor infectious ailments (Tyler, 1993). Today, chamomile is most frequently used as a sleep aid and in the promotion of wound healing. Apigenin (4, 5, 7-trihydroxyflavone) is the best studied component of chamomile. This flavonoid has inhibitory effects on adhesion molecule expression (Panes et al., 1996; Wolle et al., 1996), prostaglandin (PG) E₂ (Liang et al., 1999; Panes et al., 1996), cyclooxygenase (COX)-2 (Liang et al., 1999) production and the proinflammatory cytokine interleukin (IL)-6 (Panes et al., 1996) in cell culture models.

The herbal supplement ginseng, derived from the roots of *Panax ginseng* or *Panax quinquefolium*, is estimated to be consumed regularly by more than 6 million Americans (The Lawrence Review, 1990). Ginseng extracts are frequently consumed for their ability to enhance physical performance and for their adaptogenic effects. These effects are believed to increase the body's ability to fight stress, increase resistance to disease by strengthening normal body function and as a result, reduce the detrimental effects of the aging process (O'Hara et al., 1998; Tyler, 1993). The ginsenosides are believed to be responsible for most of ginseng's activities (Attele et al., 1999; Fleming, 1998; Tyler, 1993; Wakabayashi et al., 1997). Cell culture studies have shown that ginsenoside Rb₁, one of the primary ginsenoside components (Bae et al., 2000), can inhibit lipopolysaccharide (LPS)-induced expression of the proinflammatory cytokine tumor necrosis factor (TNF)- α (Cho et al., 1998). No in vivo data are available assessing the effect of ginseng on proinflammatory cytokine production.

Feverfew, *Tanacetum parthenium*, has been used for at least two millenia for the treatment of fever as well as headache, menstrual irregularities and stomachache (Tyler, 1993). Today, feverfew is widely used as a migraine preventative, and more recently as an aid for those suffering from arthritis (Jain & Kulkarni, 1999; Schinella et al., 1998; Williams et al., 1999). The majority of the medicinal properties are attributed to

the sesquiterpene lactone parthenolide, which inhibits expression of intercellular adhesion molecule (ICAM)-1 (Dietrich et al., 1999), COX-2 and TNF- α production (Hwang et al., 1996) in vitro. No studies have been published which evaluated the effects of parthenolide on proinflammatory cytokine production in vivo.

Lipopolysaccharide (LPS), the major biologically active component of endotoxin, is derived from cell walls of gram-negative bacteria (Roth et al., 1997). LPS is a potent inducer of inflammation and can be used to induce production of inflammatory mediators, including proinflammatory cytokines, in model systems. Induction of the proinflammatory cytokines TNF- α and IL-6 by LPS can be monitored and evaluated to study the effect of pharmacological treatment, including herbals, on the inflammatory response. In this study, we hypothesized that apigenin, ginsenoside Rb₁ and parthenolide inhibit the expression of TNF- α and IL-6 in murine macrophage cell culture and intact animal model systems. Although all three constituents inhibited proinflammatory cytokine production in vitro, these results did not adequately predict effects in vivo.

2. Materials and methods

2.1. Chemicals

All chemicals were purchased from Sigma Chemical Co. (St. Louis, MO) unless otherwise noted. For in vitro studies, apigenin, ginsenoside Rb₁ and parthenolide (Aldrich, Milwaukee, WI) were dissolved in tissue culture grade dimethyl sulfoxide (DMSO) and further diluted in Dulbecco's modified Eagle's medium (DMEM, Sigma). The final concentration of DMSO in cell culture (less than 0.01% (v/v)) was not cytotoxic as determined by the colorimetric MTT (tetrazolium) assay (Mosmann, 1983). LPS from *Salmonella typhimurium* [1.5 EU/ng LPS] was diluted in DMEM. For in vivo studies, apigenin, ginsenoside Rb₁ and parthenolide were dissolved in DMSO; LPS was dissolved in endotoxin-free tissue culture grade water.

2.2. Experimental design for in vitro studies

The murine macrophage cell line, RAW 264.7, was obtained from the American Type Culture Collection (ATCC, Rockville, MD). Cells were cultured in DMEM containing 4 mM L-glutamine, 3.7 g/l sodium bicarbonate and 4.5 g/l glucose. Medium was supplemented with 10% (v/v) fetal bovine serum (FBS) (Atlanta Biologicals, Norcross, GA), 100 U/ml penicillin, 100 μ g/ml streptomycin, 1 mM sodium pyruvate and 1% (v/v) NCTC-135 medium (Gibco BRL, Rockville, MD). Cells were maintained in a 5% CO₂ humidified incubator at 37 °C. Cell number and viability were assessed by trypan

blue dye exclusion using a hemacytometer (American Optical, Buffalo, NY).

RAW 264.7 cells (5×10^5 cells/ml) were cultured in flat-bottomed 48-well tissue culture plates (0.8 ml/well) (Corning, Corning, NY) with either apigenin, ginsenoside Rb₁ or parthenolide with or without stimulation by LPS. Supernatants were collected after 12, 24 and 48 h incubations (24 and 48 h data not shown) and stored at -80°C until analysis.

2.3. Experimental design for *in vivo* studies

All animal handling was conducted in ordinance with guidelines established by the National Institutes of Health. Experiments were designed to minimize the numbers of animals used. Female B6C3F1 mice (8–10 weeks) were obtained from Charles River (Portage, MI). Animals were housed 3–4 per cage with a 12 h light/dark cycle, provided standard rodent chow and water *ad libitum*, and acclimated to their environment as least one week before the start of experiments.

In all cases, treatment with each herbal constituent was via p.o. administration to more closely approximate human exposure through consumption of herbal dietary supplements. Mice were pre-treated for 1 h with apigenin (50 mg/kg, p.o. in 50 μl) based on efficacious results obtained by Panes et al. (1996) and Mascolo et al. (1988) using similar treatment times and doses via i.p. administration. A 2 h pre-treatment with ginsenoside Rb₁ (25 mg/kg, p.o. in 50 μl) was used based on the previously reported kinetics of ginsenoside Rb₁, and its active metabolite, ginsenoside M1 (also known as compound K) (Wakabayashi et al., 1997), and treatment dose reported as efficacious by Jun-tain et al. (1990) and Hasegawa and Uchiyama (1998). Parthenolide (50 mg/kg, p.o. in 50 μl) was administered 1 h prior to LPS based on studies by Jain and Kulkarni (1999) and Mitra et al. (2000), and dose based on consideration of studies by Jain and Kulkarni (1999) and Schinella et al. (1998). LPS was administration at 1 mg/kg, i.p. in 100 μl . Vehicle-treated mice received 50 μl DMSO, p.o. and/or 100 μl H₂O, i.p. A 90 min time point sufficient to evaluate both IL-6 and TNF- α from one blood draw was chosen for sample collection based on time course data obtained by Zhou et al. (1999). Blood was obtained by retro-orbital bleeding and allowed to clot overnight at 4°C . Serum was collected and analyzed for IL-6 and TNF- α by ELISA.

2.4. IL-6 and TNF- α determination by ELISA

Cytokine analysis for IL-6 (in vitro and in vivo) and TNF- α (in vitro) were performed using purified and biotin-conjugated rat anti-mouse IL-6 and TNF- α antibodies, respectively, from PharMingen (San Diego, CA). Streptavidin-peroxidase (Sigma) and 3,3',5,5'-tetra-

methylbenzidine (TMB, Fluka, Ronkonkoma, NY) were used for detection. Absorbance was read at 450 nm using a VmaxTM Kinetic Microplate Reader (Molecular Devices, Menlo Park, CA). For TNF- α analysis in vivo, the OptEIA Set: Mouse TNF- α (Mono/Poly) kit was used according to manufacturer's instructions.

2.5. Statistics

All statistical testing was performed using SigmaStat Statistical Analysis Software (Jandel Scientific, San Rafael, CA). For comparisons of multiple groups using parametric data, one-way analysis of variance (ANOVA) using Bonferroni's method were performed. For comparison of two groups, a Student's *t*-test was used.

3. Results

3.1. *In vitro* co-treatment with apigenin impairs LPS-induced IL-6, but not TNF- α production

To determine the potential inhibitory effects of herbal constituents on proinflammatory cytokine expression, RAW 264.7 murine macrophage cells were stimulated by LPS and co-treated with various herbal constituents. Apigenin, a flavone found in chamomile extracts and foods including parsley, bell peppers and guava was evaluated in vitro for potential inhibitory effects on LPS-induced IL-6 and TNF- α . RAW cells were co-treated with LPS and apigenin, 0.1–10 $\mu\text{g/ml}$, for 12 h. Apigenin doses evaluated were not cytotoxic as determined by the colorimetric MTT (tetrazolium) assay (results not shown) (Mosmann, 1983). LPS at 100 and 1000 ng/ml significantly induced IL-6 and TNF- α production following incubation (Fig. 1). No direct induction of proinflammatory cytokines was observed in vehicle or apigenin-only treated cells. Co-treatment with the two highest doses, 1 and 10 $\mu\text{g/ml}$ (3.7 and 37 μM , respectively), of apigenin significantly, and dose-dependently, impaired LPS-induced IL-6. Significant inhibition of LPS-induced TNF- α was not observed.

3.2. *In vitro* co-treatment with ginsenoside Rb₁ dose-dependently inhibits LPS-induced IL-6 and TNF- α production

Ginsenoside Rb₁ is one of 12 major ginsenosides derived from *Panax* species. To test ginsenoside Rb₁'s capacity to inhibit LPS-induced IL-6 and TNF- α production, RAW cells were co-treated with LPS and ginsenoside Rb₁, 0.1 to 100 $\mu\text{g/ml}$. No dose of ginsenoside Rb₁ tested caused cytotoxicity as determined by the MTT assay (results not shown). Significant production of IL-6 and TNF- α was observed following treatment

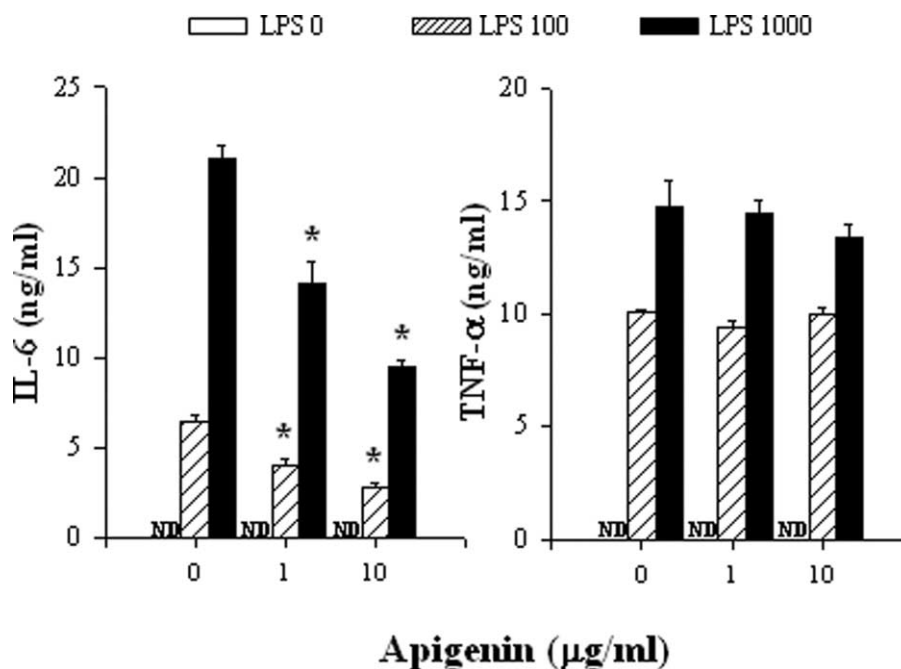


Fig. 1. Apigenin inhibits LPS-induced interleukin-6 (IL-6) but not tumor necrosis factor- α (TNF- α) production by RAW 264.7 cells. Cells (5×10^5 cells/ml) were cultured with LPS (0, 100 or 1000 ng/ml) in the presence or absence of apigenin for 12 h and supernatant analyzed for IL-6 and TNF- α . Data are mean $\pm$ SEM ($n=6$) of triplicate cultures. Values marked with an asterisk (*) differ significantly from LPS control values ($P < 0.05$). ND indicates nondetectable.

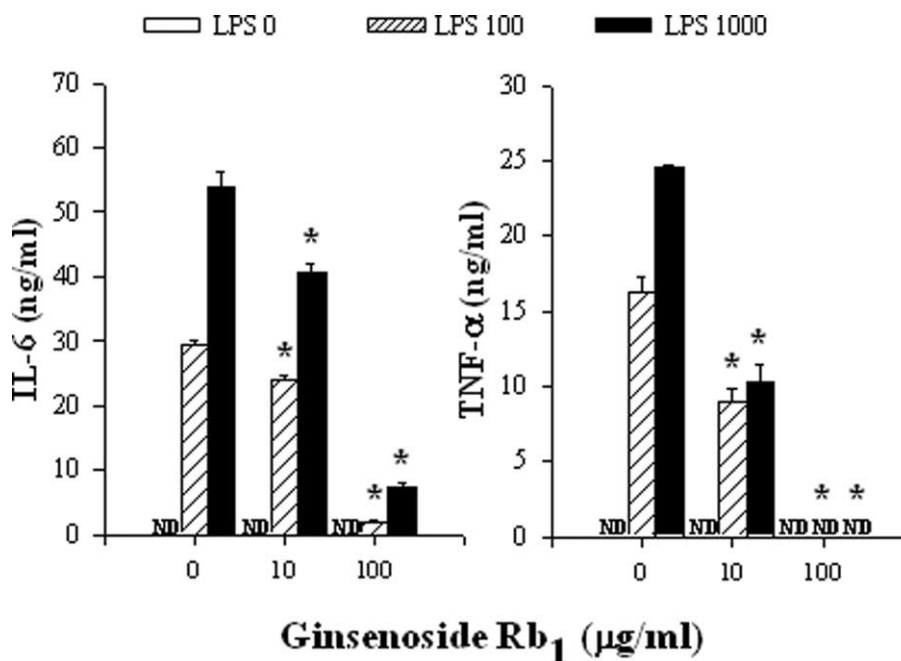


Fig. 2. Ginsenoside Rb₁ inhibits LPS-induced IL-6 and TNF- α production by RAW 264.7 cells. Cells were cultured as described in Fig. 1 with the presence or absence of ginsenoside Rb₁ and supernatant analyzed for IL-6 and TNF- α . Data are mean $\pm$ SEM ($n=6$) of triplicate cultures. Values marked with an asterisk (*) differ significantly from LPS control values ($P < 0.05$). ND indicates nondetectable.

with 100 and 1000 ng/ml LPS (Fig. 2). No direct induction of IL-6 or TNF- α was observed in vehicle or ginsenoside Rb₁-only treated cells. Ginsenoside Rb₁ co-treated cells produced significantly, and dose-dependently, less LPS-induced IL-6 following 12 h incubation. Similar

dose-dependent impairment of TNF- α was observed, with significant decreases in LPS-induced TNF- α following 10 μ g/ml (8.4 μ M) co-treatment, and a complete inhibition in cells co-treated with 100 or 1000 ng/ml LPS and 100 μ g/ml (84 μ M) of ginsenoside Rb₁.

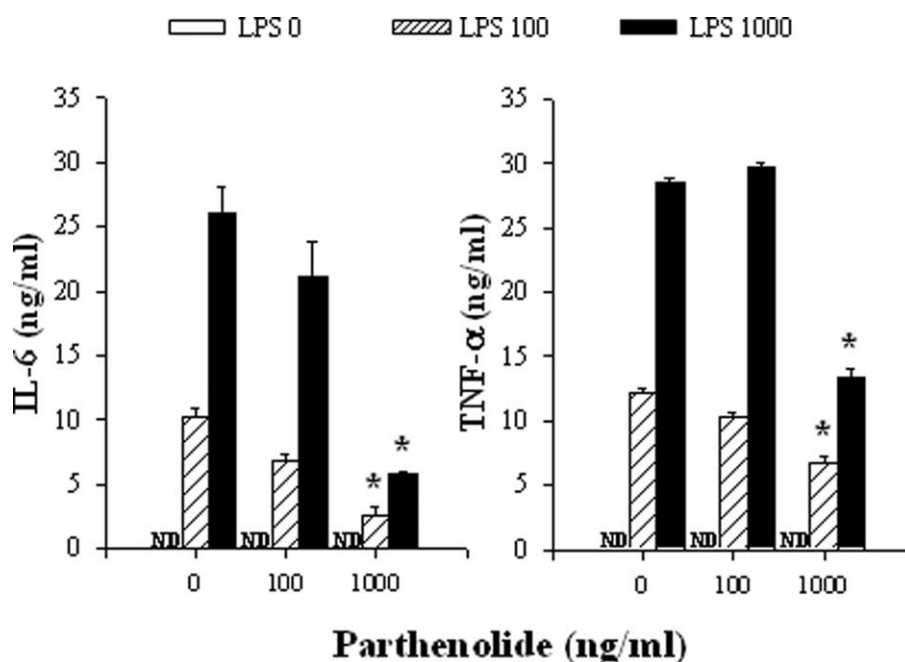


Fig. 3. Parthenolide inhibits LPS-induced IL-6 and TNF- α production by RAW 264.7 cells. Cells were cultured as described in Fig. 1 with the presence or absence of parthenolide and supernatant analyzed for IL-6 and TNF- α . Data are mean $\pm$ SEM ($n=6$) of triplicate cultures. Values marked with an asterisk (*) differ significantly from LPS control values ($P<0.05$). ND indicates nondetectable.

3.3. *In vitro* co-treatment with parthenolide inhibits LPS-induced IL-6 and TNF- α

Parthenolide, used as the constituent for standardization in commercial feverfew products, exhibits a wide variety of anti-inflammatory effects in murine and human cell systems. To determine potential effects on proinflammatory cytokine secretion, RAW cells were co-treated with LPS and 1–1000 ng/ml parthenolide. Cytotoxicity, as determined by the MTT assay, was not observed with parthenolide test doses (results not shown). Significant production of IL-6 and TNF- α was observed following 12 h exposure to LPS (Fig. 3). No direct induction of proinflammatory cytokines was observed in vehicle and parthenolide-only treated cells. Co-treatment with 1000 ng/ml (4 μ M) parthenolide significantly impaired both LPS-induced IL-6 and TNF- α .

3.4. *Apigenin pretreatment in vivo* inhibits LPS-induced IL-6 and TNF- α

In order to more closely approximate potential human effects and to compare model systems, an animal model was employed using each herbal constituent and serum accumulation of LPS-induced proinflammatory cytokines as efficacy endpoint. Female mice were pre-treated with apigenin (50 mg/kg, p.o.) based on previous *in vivo* studies by Panes et al. (1996) and Mascolo et al. (1988). After 1 h, LPS (1 mg/kg, i.p.) was administered

and blood was collected 90 min later. Analysis of the serum showed significant levels of LPS-induced IL-6 and TNF- α production in controls (Fig. 4). Serum levels of IL-6 and TNF- α were not detectable in vehicle and apigenin control animals. One hour pretreatment with apigenin caused a significant reduction in LPS-induced proinflammatory cytokine production. LPS-induced IL-6 (Fig. 4A) was significantly reduced (35%), while TNF- α (Fig. 4B) was significantly reduced (33%) compared to animals treated with vehicle and LPS alone.

3.5. *Ginsenoside Rb₁ pretreatment in vivo* selectively inhibits LPS-induced TNF- α

The *in vivo* effects of ginsenoside Rb₁ on LPS-induced IL-6 and TNF- α levels in serum were also evaluated. Ginsenoside Rb₁ (25 mg/kg, p.o.) was administered 2 h prior to LPS (1 mg/kg, i.p.) based on the previously reported kinetics of ginsenoside Rb₁, and its active metabolite, ginsenoside M1 or compound K, in the sera of treated mice (Wakabayashi et al., 1997). Analysis of the serum following 90 min exposure to LPS showed significant induction of IL-6 and TNF- α , whereas these cytokines were unaffected in vehicle and ginsenoside Rb₁ control-treated animals (Fig. 5). No changes in IL-6 production (Fig. 5A) were observed with ginsenoside Rb₁ pretreatment. However, TNF- α production was significantly inhibited (Fig. 5B), (37%), by ginsenoside Rb₁ pretreatment.

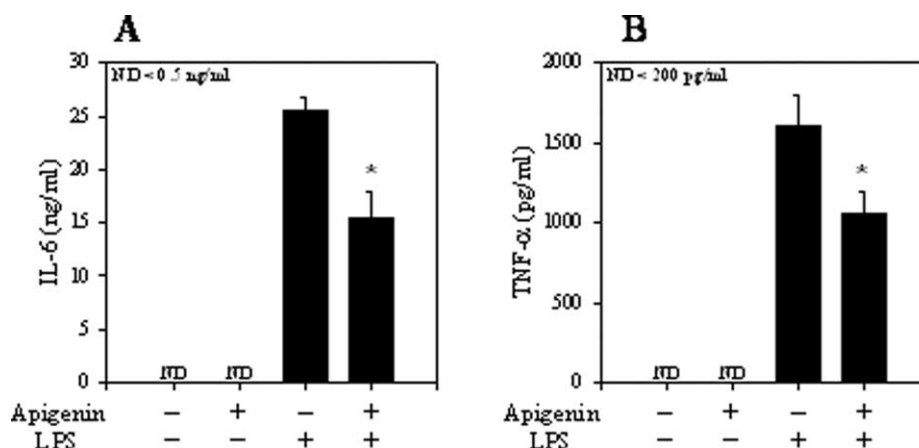


Fig. 4. Apigenin pretreatment in vivo inhibits LPS-induced IL-6 and TNF- α production. Female B6C3F1 mice were pretreated with apigenin (50 mg/kg, gavage) or vehicle (50 μ l DMSO) for 1 h. LPS (1 mg/kg, i.p.) or vehicle (100 μ l sterile water) was administered, and after 90 min blood was collected and serum analyzed for (A) IL-6 and (B) TNF- α . Data are mean $\pm$ SEM ($n=12$, controls $n=6$). Values marked with an asterisk (*) differ significantly from LPS control values ($P<0.05$). ND indicates nondetectable. Data are a combination of three separate experiments.

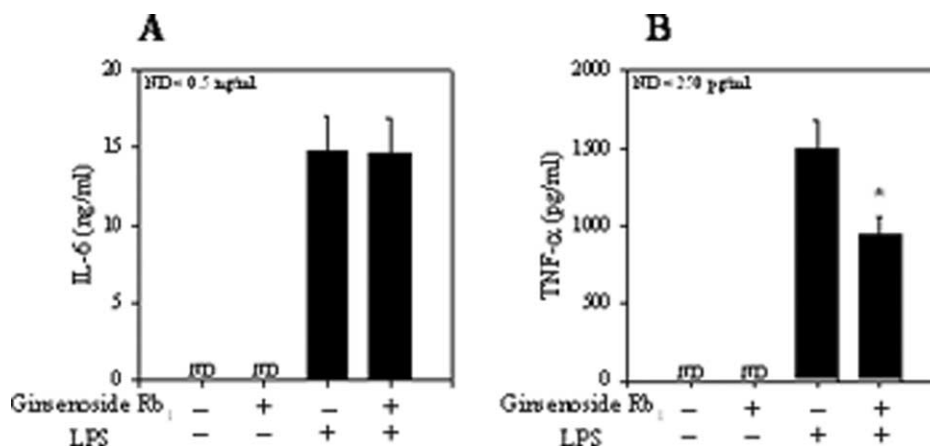


Fig. 5. Ginsenoside Rb₁ pretreatment in vivo inhibits LPS-induced TNF- α but not IL-6 production. Female B6C3F1 mice were pretreated with ginsenoside Rb₁ (25 mg/kg, gavage) or vehicle (50 μ l DMSO) for 2 h. LPS (1 mg/kg, i.p.) or vehicle (100 μ l sterile water) was administered, and after 90 min blood was collected and serum analyzed for (A) IL-6 and (B) TNF- α . Data are mean $\pm$ SEM ($n=12$, controls $n=6$). Values marked with an asterisk (*) differ significantly from LPS control values ($P<0.05$). ND indicates nondetectable. Data are a combination of three separate experiments.

3.6. Parthenolide pretreatment in vivo shows no inhibitory effects on LPS-induced IL-6 and TNF- α

Parthenolide's ability to impair LPS-induced IL-6 and TNF- α was also evaluated in vivo. Mice were pretreated with parthenolide (50 mg/kg, p.o.) for 1 h prior to LPS (1 mg/kg, i.p.) administration. Serum was analyzed for IL-6 and TNF- α following a 90 min exposure to LPS. Serum levels of IL-6 and TNF- α were below detection in vehicle and parthenolide control-treated animals (Fig. 6). Although there was significant production of IL-6 and TNF- α by LPS, parthenolide pretreatment caused no significant changes in cytokine production for either IL-6 (Fig. 6A) or TNF- α (Fig. 6B).

4. Discussion

Consumer use of dietary supplements has grown significantly since the passage of the DSHEA in 1994 and herbals account for a considerable portion of total supplement sales (Blendon et al., 2001). Despite this growth, (Commission on Dietary Supplement Labels, 1998; Sarubin, 2000) there is still a considerable lack of scientific evidence to clearly support efficacy and, in some cases, safety of these products. The objective of this study was to establish a cell culture system that could be used to evaluate the potential efficacy of herbal constituents with presumed anti-inflammatory properties, and to determine if the results of cell culture studies

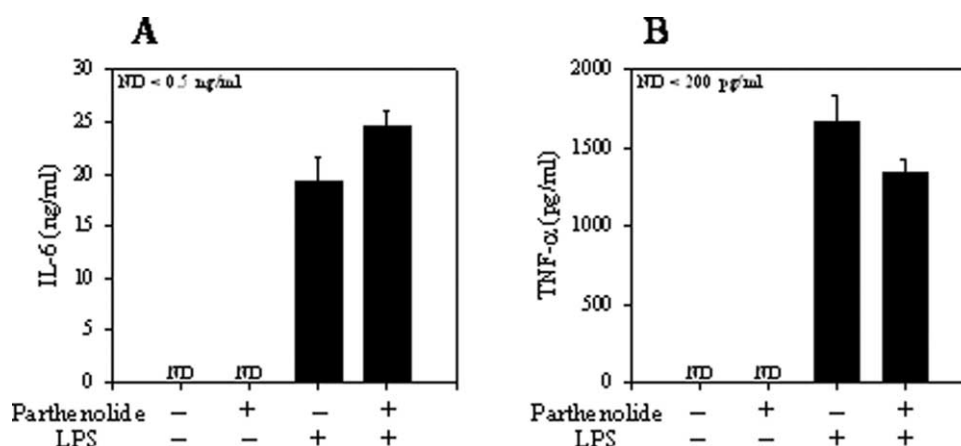


Fig. 6. Parthenolide pretreatment in vivo does not inhibit LPS-induced IL-6 and TNF- α production. Female B6C3F1 mice were pretreated with parthenolide (50 mg/kg, gavage) or vehicle (50 μ l DMSO) for 1 h. LPS (1 mg/kg, i.p.) or vehicle (100 μ l sterile water) was administered, and after 90 minutes blood was collected and serum analyzed for (A) IL-6 and (B) TNF- α . Data are mean $\pm$ SEM ($n = 12$, controls $n = 6$). ND indicates non-detectable. Data are a combination of three separate experiments.

could accurately predict effects in an animal model. The potential anti-inflammatory properties of apigenin, ginsenoside Rb₁ and parthenolide were evaluated in RAW 264.7 cells because this clonal macrophage model produces high concentrations of IL-6 and TNF- α in culture upon activation with LPS, and reduced the need for additional animals required to obtain and use primary cultures. Cytokine levels were also evaluated in mice using LPS as a stimulant. The cytokines IL-6 and TNF- α were evaluated because these proinflammatory cytokines play a key role in the inflammatory response and can be easily quantified in supernatant and serum samples. Studies of apigenin's effect on proinflammatory cytokines are limited. The results of this in vitro study showed that apigenin significantly inhibited LPS-induced IL-6, but not TNF- α . Gerritsen et al. (1995) stimulated human endothelial cells with TNF- α and found that apigenin co-treatment inhibited IL-6 production as was observed herein for LPS treatment. However, in another study evaluating the inhibitory effects of apigenin on TNF- α production, Mastuda et al. (2002) showed that apigenin, at an IC₅₀ = 5.3 μ M, inhibited antigen-IgE-mediated TNF- α secretion in RBL-2H3 mast cells. A potential reason for the differences in results may be the differences in stimuli and cell types used.

Although significant reductions in LPS-induced IL-6 by apigenin were observed here in cell culture, both IL-6 and TNF- α accumulation were inhibited in the animal model. No previous studies have evaluated the effect of apigenin on proinflammatory cytokine levels in vivo. Engelmann et al. (2002) assessed apigenin's in vitro and in vivo effects on cytostatic and anti-angiogenic action. The results showed effects in vitro, but lack of efficacy in vivo. However, animal studies evaluating the effect of apigenin on TNF-induced ICAM-1 expression (Panés et al., 1996) and carrageenan-induced PGE₂ and leukotriene

(LT)B₄ (Mascolo et al., 1988) revealed inhibitory effects of apigenin similar to those observed in this study. The Panes study employed a 100 mg/kg dose and administered the apigenin i.p. 1 h prior to and 2 h following TNF-stimulation, while the Mascolo study employed a single 24 mg/kg, i.p. dose administered 15 min prior to the stimulus. The doses differed from the single 50 mg/kg, p.o. pretreatment employed in the present study, which was chosen as a result of observed anti-inflammatory effects in the above named studies. It is not clear how differences in mode of administration impact the results based on the specific doses evaluated in each of these studies since no evaluation of the pharmacokinetics of apigenin in mice are available. Additionally, because no reported levels of apigenin in chamomile products are available, it is unclear how the 50 mg/kg dose used in this study compares to reasonable human consumption through chamomile extracts.

Although ginsenoside compounds, including Rb₁, have been studied frequently relative to their metabolism (Akao et al., 1998; Attele et al., 1999; Chen et al., 1980; Hasegawa & Uchiyama, 1998; Wakabayashi et al., 1997), relatively few studies have evaluated the effect of ginsenoside Rb₁ on proinflammatory cytokine production in vitro and in vivo. The results of those latter studies are generally consistent with the findings herein. For example, Cho et al. (2001) showed that ginsenoside Rb₁ at an IC₅₀ of 56.5 μ M significantly impaired LPS-induced TNF- α in mouse and human macrophage cells. In another study that investigated the immunostimulatory activity of *Panax quinquefolius* (American ginseng) extracts containing ginsenoside compounds but no polysaccharides, significant TNF- α stimulating activity was observed (Assinewe et al., 2002). This effect, however, could not be induced using pure ginsenoside Rb₁. This finding agrees with our results showing that ginsenoside Rb₁ alone did not affect TNF- α levels in culture

supernatants and suggests that the immunostimulatory properties of the ginseng extract were not due to the presence of ginsenoside Rb₁, generally the ginsenoside in highest abundance in *P. quinquefolius* extracts (Harkey et al., 2001).

Although our in vitro data showed significant inhibitory effects of ginsenoside Rb₁ on LPS-induced IL-6 and TNF- α , our in vivo data did not follow the same pattern. Rather, oral gavage with ginsenoside Rb₁ inhibited LPS-induced TNF- α production but not IL-6 levels. Other studies evaluating ginsenosides in vivo support the anti-inflammatory claims attributed to ginsenoside Rb₁. A model of chronic inflammation in aged rats induced with multiple daily injections of LPS was used by Yu and Li (2000) to study the effect of ginseng root saponins (ginsenosides) on IL-1 β and IL-6 mRNA expression. Daily treatment with 10, 20 or 40 mg/kg, i.p. ginseng root saponins markedly decreased the mRNA expression levels of both proinflammatory cytokines in the hippocampus compared to LPS-stimulated animals which did not receive ginseng root saponin treatment. These findings of IL-6-related effects contrast with our model, possibly because of the different route of exposure, target tissue and endpoint.

Compared to apigenin and ginsenoside Rb₁, many more studies have been undertaken to elucidate the anti-inflammatory properties of feverfew extract and parthenolide. For example, Hwang et al. (1996) showed LPS-stimulated TNF- α synthesis is inhibited by parthenolide in alveolar macrophage cells. Similarly, Uchi et al. (2002) showed that parthenolide impairs LPS-induced TNF- α production in human monocyte-derived dendritic cells. Additionally, parthenolide pre-incubation of HeLa cells prevents the induction of transcription from the IL-6 promoter (Bork et al., 1997). The results of previous in vitro studies are similar to our findings that parthenolide significantly impaired LPS-induced IL-6 and TNF- α production.

Although in vitro results showed here that parthenolide inhibited LPS-induced IL-6 and TNF- α , the in vivo findings were not in agreement. Only a limited amount of in vivo data are available for parthenolide, and specifically, no studies have been found in the literature which have evaluated its effects on proinflammatory cytokine responses in animals. However, other in vivo anti-inflammatory endpoints have been evaluated with respect to feverfew extracts and parthenolide in mice and rats (Jain & Kulkarni, 1999). When feverfew extracts from *Tanacetum parthenium* were administered by oral gavage, or pure parthenolide was injected i.p., both anti-inflammatory and antinociceptive effects were observed. Similarly, Tournier et al. (1999) showed that feverfew extracts and parthenolide from *Tanacetum vulgare*, administered by oral gavage, significantly reduced gastric ulcer index induced by ethanol in rats. Although these findings support the anti-inflammatory

properties of parthenolide in animals, the anti-inflammatory effects could be the result of inhibition of multiple aspects of the inflammatory response. Other proinflammatory mediators including chemokines (MIP-2), plasma enzyme mediators (complement, kinin and clotting systems) and lipid mediators (COX, PG, platelet activating factor), may be affected rather than proinflammatory cytokines. This may explain the lack of effect observed here on the reduction of LPS-induced proinflammatory cytokine levels in the serum of parthenolide pretreated animals when compared to other anti-inflammatory studies using parthenolide.

In these studies, specific herbal constituents that are believed to be responsible for a majority of the extract's activity were chosen for evaluation. This was done to minimize variability that is frequently encountered in whole herb extract products (Harkey et al., 2001; Nelson et al., 2002). The intention was to use the results as a potential predictor of effects of whole herb extracts. Groenewegen and Heptinstall (1990), in a comparison of parthenolide and whole feverfew extracts on human platelet activity, showed that there are great similarities of feverfew extract and parthenolide on platelet aggregation and concluded, in platelets, that parthenolide from feverfew was responsible for the biological effects of the extract. In vivo studies by Tournier et al. (1999) showed similar effectiveness of parthenolide in comparison to whole extracts of feverfew in the cytoprotection of gastric mucosa against alcohol injury. However, Mitra et al. (2000) showed that parthenolide alone inhibited 5-hydroxytryptamine, but that it was evident the feverfew plant extract, which contains many other compounds, was more potent than parthenolide alone. These studies showed that focusing on specific individual constituents derived from herbal extracts may be used to predict the potential effect of a whole herb extract, but that additional, confirmatory studies using whole extracts need to be completed. These approaches may be used as a potential means of standardization of activities for parthenolide in feverfew and ginsenosides in ginseng.

Overall, the results of this study showed the compounds tested could inhibit proinflammatory cytokine responses. However, cell culture data did not always accurately predict the results in an animal model. This was supported by the findings that none of the three compounds evaluated displayed similar inhibitory activity in the cell and animal model systems for both IL-6 and TNF- α production levels. Discrepancies between cell culture data and animal data are not entirely surprising considering only one cell type exists in the culture studies, whereas multiple cell types capable of producing proinflammatory cytokines are present in a whole animal system. Additionally, other inflammatory mediators may influence the cytokine response. Finally, other key factors are metabolism and

distribution of the herbal constituents. Future perspectives should include understanding the molecular basis for inhibitory effects of these specific herbal constituents on proinflammatory cytokine gene expression and specifically the role of altered signal transduction.

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